

04-DNS-VISIBILITY-GAP

DNS Visibility Gap & Monitoring Limitations

Detection Scenario

This lab documents a visibility gap analysis conducted within a simulated security operations center (SOC) environment. An attempt was made to identify potential dynamic domain generation algorithms (DGA), DNS tunneling, and Command and Control (C2) traffic initiated from a Linux endpoint. However, the default operating system architecture creates a complete blind spot for native DNS query resolution logging within traditional system logs (syslog), presenting an immediate detection gap.

Objective

Evaluate and document the technical limitations of default DNS logging on Linux endpoints using systemd-resolved, validate SIEM detection gaps when querying standard logs, and propose secondary compensation controls.

Environment

- **Victim Server:** Ubuntu Server (Default logging profile)
- **SIEM Platform:** Splunk Enterprise
- **Auditing Daemon:** system-resolved
- **Log Shipper:** Splunk Universal Forwarder

Attack Simulation (Attempted Generation)

To test if malicious domain activity could be audited from standard operating system telemetry, a script was executed on the victim host to generate 100 sequential queries mimicking a Domain Generation Algorithm (DGA) or DNS data exfiltration attempt:

```
for i in {1..100}; do dig random$i.malicious-domain.com; done
```

Note: Although the queries were successfully processed by the network interface, the telemetry generation inside the host logs did not record the targeted domains (malicious-domain.com) or query details due to service configuration limits

Data Source

Log File	/var/log/syslog
Index	Main
Sourcetype	syslog

Detection Logic & SPL Queries

1. Naive DNS Query Search (Búsqueda inicial infructuosa)

Query ejecutada con el propósito de localizar trazas textuales de consultas en la fuente general del sistema:

```
index=main sourcetype=syslog "query"
```

2. Contextual Telemetry Analysis (Triage de eventos capturados)

Query avanzada diseñada para aislar la cadena de texto y verificar el origen real de la palabra "query" dentro de syslog, demostrando que los hits corresponden a ruido operativo y no a resolución de red:

```
index=main sourcetype=syslog "query"
| table _time host process _raw
```

Analysis

The initial security search yielded multiple telemetry events containing the string "query". However, contextual validation and tactical triage confirmed that these entries originated from the Linux kernel and ACPI power management events (e.g., system queries regarding hardware status) rather than actual network DNS query actions.

By default, systemd-resolved acts as a local stub resolver but does not output a continuous audit stream of queries and answers to syslog or journald unless high-level debug logging is manually injected. Consequently, there is an absolute monitoring gap regarding domain-resolution activity within the centralized SIEM.

Findings

- **False Positive Identification:** Events matching the literal string "query" were categorized as hardware/kernel noise rather than network transactions.
- **Telemetry Blind Spot:** The default local resolver failed to generate security-relevant transaction auditing for lookups.
- **DGA/Tunneling Invisibility:** Malicious sequential bursts (dig random\$.i...) left no trace within /var/log/syslog.
- **SIEM Collection Limitations:** Universal Forwarder data structures confirmed that standard system inputs are blind to transient DNS data unless explicit service engines are deployed.

MITRE ATT&CK Mapping

- **Status:** Detection Gap / Visibility Limitation
- **Impacted Matrix:** Prevents defense validation against **T1071.004 (Application Layer Protocol: DNS)**, **T1568 (Dynamic Resolution)**, and **T1048 (Exfiltration Over Alternative Protocol)**.

Impact

The lack of structural DNS visibility severely restricts a security team's ability to intercept active command-and-control beacons, isolate automated malware families running DGAs, or identify data exfiltration staging through covert DNS tunneling protocols.

Classification

False Positive (Regarding DNS activity detection; verified as a **Visibility Gap**).

Recommendations

1. Deploy robust, security-focused local logging engines such as dnsmasq or configure authoritative/recursive query log modules in BIND to capture real-time resolution telemetry.
2. Modify the systemd-resolved verbosity settings to active query tracking or monitor the loopback resolution interface (127.0.0.53) using dedicated packet capture tools.
3. Integrate network-level DNS logging tools (such as Zeek/Bro or CoreDNS) directly into the Splunk Universal Forwarder ecosystem.
4. Build real-time correlation alerts between high-frequency execve shell tools (e.g., explicit usage of dig or nslookup via auditd) and external threat intelligence indicators.

Screenshots Required

1. **Failed DNS visibility searches:** Pantallazo en Splunk mostrando que la consulta `index=main sourcetype=syslog "query"` devuelve logs irrelevantes del kernel/ACPI en lugar de dominios.

New Search Save As Create Table View Close

`index=main sourcetype="syslog" "query"` Last 24 hours Q

✓ 2 events (3/24/26 12:00:00.000 AM to 3/25/26 12:42:16.000 AM) No Event Sampling Job || ■ → 📄 ⬇ 🔍 Smart Mode

Events (2) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection × Deselect 1 hour per column

Format Show: 20 Per Page View: List

	i	Time	Event
>		3/24/26 11:25:22.646 PM	2026-03-24T23:25:22.646809-03:00 ubuntu-server kernel: ACPI: [Firmware Bug]: BIOS _OSI(Linux) query ignored host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/24/26 7:16:42.414 PM	2026-03-24T23:16:42.414048-03:00 ubuntu-server kernel: ACPI: [Firmware Bug]: BIOS _OSI(Linux) query ignored host = ubuntu-server source = /var/log/syslog sourcetype = syslog

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
date_hour 2

New Search Save As Create Table View Close

`index=main sourcetype="syslog"` 30 minute window Q

46 of 47 events matched No Event Sampling Job || ■ → 📄 ⬇ 🔍 Smart Mode

Events (46) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection × Deselect 1 minute per column

Format Show: 20 Per Page View: List < Prev 1 2 3 Next >

	i	Time	Event
>		3/25/26 12:35:01.814 AM	2026-03-25T00:35:01.814743-03:00 ubuntu-server CRON[2984]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1) host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/25/26 12:33:10.601 AM	2026-03-25T00:33:10.601647-03:00 ubuntu-server systemd[1]: packagekit.service: Deactivated successfully. host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/25/26 12:33:10.599 AM	2026-03-25T00:33:10.599361-03:00 ubuntu-server PackageKit: daemon quit host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/25/26 12:30:17.618 AM	2026-03-25T00:30:17.618685-03:00 ubuntu-server systemd[1]: Finished sysstat-collect.service - system activity accounting tool. host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/25/26 12:30:17.618 AM	2026-03-25T00:30:17.618641-03:00 ubuntu-server systemd[1]: sysstat-collect.service: Deactivated successfully. host = ubuntu-server source = /var/log/syslog sourcetype = syslog
>		3/25/26	2026-03-25T00:30:17.614691-03:00 ubuntu-server systemd[1]: Starting sysstat-collect.service - system a

SELECTED FIELDS
a host 1
a source 1
a sourcetype 1

INTERESTING FIELDS
a apparmor 1
a class 1
a comm 2
date_hour 1
date_mday 1
date_minute 10
date_month 1
date_second 9
a date_wday 1
date_year 1
date_zone 1
a denied_mask 1
fault 1